

Independent Events

Two events, A and B , are said to be **independent** if

$$P(A \cap B) = P(A)P(B).$$

Corollary

If events A and B are independent and $P(B) > 0$, then

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = P(A).$$

Meaning: The probability of event A occurring given event B does not depend on probability event B .

Exercise (!) If events A and B are independent, then events
(i) A and B' , (ii) A' and B , (iii) A' and B' are independent.

Mutually Independent Events

Events $A_1, A_2, \dots, A_n$ are mutually independent if for any subcollection $A_{j_1}, A_{j_2}, \dots, A_{j_m}$, i.e. $j_i \in \{1, 2, \dots, n\}$ for all $1 \leq i \leq m$

$$P(A_{j_1} \cap A_{j_2} \cap \dots \cap A_{j_m}) = P(A_{j_1})P(A_{j_2}) \dots P(A_{j_m}).$$

Pairwise Independent Events

Events $A_1, A_2, \dots, A_n$ are pairwise independent if for events A_i and A_j , $i \neq j$, $P(A_i \cap A_j) = P(A_i)P(A_j)$ or A_i and A_j are independent events.

- Mutual Independence $\implies$ Pairwise Independence.
- Pairwise Independence $\not\implies$ Mutual Independence.

A random variable is a function

$$X : \Omega \rightarrow \mathbb{R}.$$

A discrete random variable is a random variable that can only take a **finite** or at most **countably infinite** (denumerable) number of values.

Probability Mass Function (pmf)

Given a **discrete random variable** X taking values $x_1, x_2, \dots$, its **probability mass function** or the **frequency function** of a random variable X is the function p where

$$p(x_i) = P\{X = x_i\}.$$

Note that $\sum_{i=1}^{\infty} p(x_i) = 1$.

Cumulative Distribution Function (cdf)

The cumulative distribution function (cdf) of a random variable is defined as the function $F : \mathbb{R} \rightarrow [0, 1]$ where

$$F(x) = P\{X \leq x\}, \quad -\infty < x < \infty.$$

- F is always increasing, right-continuous and it always holds that

$$\lim_{x \rightarrow -\infty} F(x) = 0 \text{ and } \lim_{x \rightarrow \infty} F(x) = 1.$$

- Remark: Often in discrete random variables, $F(x)$ is NOT a continuous function everywhere.

- $P\{a < X \leq b\} = P\{X \leq b\} - P\{X \leq a\} = F(b) - F(a).$

- $P\{X < a\} = \lim_{x \rightarrow a^-} F(x)$

- $P\{a \leq X \leq b\} = P\{X \leq b\} - P\{X < a\} = F(b) - \lim_{x \rightarrow a^-} F(x)$

Tip: Draw a number line.

Expectation of a random variable can be seen as the **mean** or μ .

Expectation, $E(X)$

If X is a **discrete random variable** with frequency function $p(x)$, the expected value of X , denoted by $\mathbb{E}(X)$, is

$$\mu_X = \mathbb{E}(X) = \sum_i x_i p(x_i) \text{ provided } \sum_i |x_i| p(x_i) < \infty.$$

Expectation of a function, $E(g(X))$

Let X be a discrete random variable of pdf $p(\cdot)$ and $Y = g(X)$. Then

$$\mathbb{E}(g(X)) = \sum_x g(x) p(x) \text{ provided } \sum_x |g(x)| p(x) < \infty.$$

Examples

$$\mathbb{E}(X^2) = \sum_x x^2 p(x), \quad \mathbb{E}(X^n) = \sum_x x^n p(x).$$

Properties of Expectations

- $\mathbb{E}(a) = a$, for all constants a .
- $\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$, for all constants a and b .

Variance, $\text{Var}(X)$

If X is a random variable with expected value μ_X , the variance of X is

$$\text{Var}(X) = \mathbb{E}([X - \mu_X]^2)$$

if the expectation exists. Standard deviation of $X = \sqrt{\text{Var}(X)}$.

Properties of Variance

- $\text{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2$ or $\mathbb{E}(X^2) - [\mu_X]^2$.
- $\text{Var}(aX + b) = a^2\text{Var}(X)$ for all constants a, b .

Bernoulli Random Variable

A **Bernoulli random variable** takes on only two values: 1 and 0, with probabilities p and $1 - p$, respectively for a single trial. In other words,

$$p(1) = p \quad p(0) = 1 - p.$$

Binomial Distribution

Given n **independent** trials, $P(\text{success}) = p$, $P(\text{failure}) = 1 - p$.

X is a **binomial random variable** with parameters n and p , denoted by $X \sim \text{Bin}(n, p)$. The probability of having k successes is

$$p(k) = P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

Idea: Binomial distribution can be seen as n independent Bernoulli Random Variables.